1 Introduction and inequalities

Week 1 · Week summary

Core ideas, lesson by lesson

1 Equations: degree, type, and roots

- Degree = highest power after expanding, moving everything to one side, and simplifying.
- 1st degree $ax + b = 0$ (one root), 2nd $ax^2 + bx + c = 0$ (up to two), 3rd up to three: degree n, at most n roots.
- Zero product rule: $A \cdot B = 0$ means $A = 0$ or $B = 0$. Check each root by substitution.

2 Second-degree equations and the parabola

- $x = (-b \pm \sqrt{\Delta})/2a$ with $\Delta = b^2 - 4ac$: $\Delta > 0$ two roots, $\Delta = 0$ one, $\Delta < 0$ none.
- $a > 0$ opens up (minimum); $a < 0$ opens down (maximum).
- Vertex $V = (-b/2a, -\Delta/4a)$; axis $x = -b/2a$; y-intercept c; x-intercepts are the roots.

3 Radical, absolute value, and fraction equations

- Radicals: isolate, square, then check every answer.
- $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$; $|A| = k \rightarrow A = k$ or $A = -k$.
- Fractions: write the forbidden values first, then multiply by the denominator.

4 From words to equations

- Name the unknown with its unit before writing anything else.
- Translate sentence by sentence; check the answer in the story.
- Discard answers that make no sense: negative lengths, fractions of people.

5 Higher-degree polynomials: Ruffini's rule

- Candidates: ± divisors of the constant (over divisors of the leading coefficient).
- Remainder 0 means a root; the bottom row is the quotient.
- Write 0 for missing powers, and repeat until a quadratic is left.

6 Types of intervals and linear inequalities

- Bounded: (a, b), [a, b], [a, b), (a, b]. Infinite: (a, ∞), [a, ∞), (−∞, a), (−∞, a].
- [ ] and filled dot = included; ( ) and hollow dot = excluded; ∞ always takes ( ).
- Multiplying or dividing by a negative flips the inequality.

7 Polynomial and rational inequalities: sign tables

- Zero on one side, factor, critical points, one test value per interval.
- ≤ or ≥ include the numerator's zeros; denominator zeros are always excluded.
- For a quadratic, the sign table is the parabola read along the x-axis.

8 Systems, absolute values by cases, and two variables

- A system keeps the intersection of the solutions.
- Two absolute values: split the line where each inside is zero, solve each case, keep answers inside their case.
- In two variables: ≤ ≥ solid boundary, < > dashed; test a point to pick the side.

Interval	$x < -4$	$-4 < x < 3$	$x > 3$
Test value	$x = -5$	$x = 0$	$x = 4$
$x + 4$	−	+	+
$x - 3$	−	−	+
$\mathbf{(x + 4)(x - 3)}$	+	−	+

> 0



Watch out

Mistake	Fix
Forgetting to flip the sign when dividing by a negative number.	Circle every negative divisor before dividing.
Including a zero of the denominator in the answer.	Mark those points with a hollow dot from the start.
Reading the degree before simplifying.	Expand and cancel first; only then look at the highest power.
Dividing both sides by x and losing the root $x = 0$.	Factor x out instead: $x^2 = 5x \rightarrow x(x - 5) = 0 \rightarrow x = 0$ or $x = 5$.
Simplifying $(4 \pm 2\sqrt{2})/2$ as $4 \pm \sqrt{2}$.	Divide every term: $(4 \pm 2\sqrt{2})/2 = 2 \pm \sqrt{2}$.
Writing the vertex as $x = b/2a$.	Keep the minus sign: $x = -b/2a$. Check: the vertex is halfway between the roots.
Keeping $x = -1$ in $\sqrt{(x + 2)} = x$.	A square root is never negative: $\sqrt{1} = 1 \neq -1$, so $x = -1$ is extraneous.
Skipping a missing power in the table: writing 1 -7 -6 for $x^3 - 7x - 6$.	Write a 0 for every missing power: 1 0 -7 -6.